

Dealer Performance Benchmarking

Analytics Variable Proposal

Pentana Vehicle Sales Data · Australia

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Purpose of this document: This proposal outlines the set of analytics variables that can be computed to benchmark automotive dealer performance across Australia. Variables are grouped by business dimension and assessed for feasibility. The data source is the Pentana vehicle sales dataset covering approximately **4.9 million retail VINs** across **2,034 dealer proxy keys** (each key = one brand at one postcode).

Feasibility legend used in all tables:

Ready	Computable from existing data at full or near-full coverage
Directional	Computable from existing data but on a subset of records (~30–65%); suitable for directional analysis and relative comparison
Ext. data	Requires one free or low-cost external dataset (e.g. VFACTS, ABS, RRP table)
Future	Requires a new data source not currently in the pipeline (F&I system, inventory feed, service history)

Why Dealer Benchmarking?

Dealers vary significantly in volume, operational speed, vehicle mix, and pricing. Without benchmarking, a dealer has no way to know whether their 90-day order-to-delivery time is exceptional or below average for their brand — and the OEM has no systematic way to identify which dealers need support.

This proposal covers **eight dimensions** of dealer performance that can be computed from Pentana vehicle sales data, enriched where beneficial with free external sources. A final section outlines variables that become available once additional data sources (insurance, inventory, finance) are connected.

1. Sales Volume & Growth

Understanding whether a dealer is growing, stable, or declining — and how they rank among peers selling the same brand.

Variable	What it measures	Derived from	Feasibility
Total units sold	Annual vehicle deliveries per dealer per brand	Delivery records	Ready
Units sold — last 12 months	Rolling active-dealer volume; filters out discontinued brands	Delivery date	Ready
Year-on-year volume growth	(Current year / prior year) – 1. Identifies growing vs declining dealers	Delivery date by year	Ready
Rolling 3-month units	Short-term momentum signal; detects sudden drops or spikes	Delivery date	Ready
Volume rank within brand	Percentile rank of dealer vs all dealers selling the same brand nationally	Delivery records	Ready
Volume trend direction	Binary flag: growing or shrinking vs prior 12 months	Delivery date	Ready
Peak-month concentration	Share of annual sales in top 2 calendar months. High value indicates end-of-quarter incentive-driven sales culture rather than organic demand	Delivery date by month	Ready

2. Sales Channel Mix

How each dealer's volume splits across sales channels. Fleet-heavy dealers behave very differently from retail-first dealers in terms of margins, inventory cycles, and customer relationships.

Variable	What it measures	Derived from	Feasibility
Retail mix %	Share of sales to private buyers	Sale type field	Directional
Fleet mix %	Share of sales to corporate fleet accounts	Sale type field	Directional
Government mix %	Share of sales to government/public sector	Sale type field	Directional
Rental mix %	Share of sales to rental companies. Rental-heavy dealers often operate on thinner margins and differ in stock management	Sale type field	Directional
Fleet concentration flag	Binary: dealer is fleet-dominant (fleet >40% of sales). Useful for separate benchmarking peer groups	Derived from fleet mix	Directional

Note: sale type field is populated for approximately 65% of records; results are directional and should be compared within brand peer groups.

3. Vehicle Portfolio & Segment Mix

What types of vehicles a dealer sells — SUVs, utes, passenger cars, commercial vans — and whether they are moving current-year stock or clearing older inventory.

Variable	What it measures	Derived from	Feasibility
SUV / wagon share %	Share of high-margin SUV and wagon segment in total sales	Body style (VIN-enriched)	Ready
Ute / cab chassis share %	Share of utility and cab chassis — commercial vs personal buyer indicator	Body style	Ready
Passenger car share %	Share of sedan and hatchback. Declining nationally; useful as a portfolio transition signal	Body style	Ready
Van share %	Commercial van concentration — signals trade and business buyer focus	Body style	Ready
Portfolio diversity index	Count of distinct body styles sold. Breadth of portfolio offering per dealer	Body style	Ready
AWD / 4WD share %	Share of all-wheel and four-wheel drive variants. Signals adventure/commercial vs urban buyer base	Drive type field	Directional
Average model age at delivery	Average gap between vehicle model year and delivery year. High values indicate a dealer is clearing older stock, typically with heavier discounting	Model year + delivery date	Ready
Aged stock rate %	Proportion of vehicles where model year is two or more years behind delivery year. A 20%+ rate is a significant discounting risk signal	Model year + delivery date	Ready
Premium brand flag	Binary indicator for premium/luxury brand dealers (BMW, Mercedes-Benz, Audi, Lexus, Porsche, Genesis, Volvo). Enables separate peer group benchmarking	Brand name	Ready
Premium variant mix %	Share of high-specification variants vs base variants within the same model. Indicates upsell effectiveness	Vehicle variant + RRP table	Ext. data

4. Electrification Transition

How far along each dealer is in transitioning their sales mix toward electric and electrified vehicles — a key strategic dimension for both OEMs and dealers.

Variable	What it measures	Derived from	Feasibility
EV units sold	Count of battery electric vehicles delivered	Fuel type (VIN-enriched)	Ready
EV share %	EV units as a proportion of total sales. Benchmarks electrification progress vs brand peers	Fuel type	Ready
EV rank within brand	Percentile rank of dealer's EV share vs all dealers for the same brand	Derived	Ready

Variable	What it measures	Derived from	Feasibility
EV year-on-year growth	Growth rate of EV unit sales. Identifies dealers leading or lagging the brand's electrification push	Delivery date + fuel type	Ready
EV + hybrid combined share %	Total electrified share (BEV + PHEV + HEV). Full picture of transition	Fuel type	Directional

Note: fuel type enrichment is available for approximately 44% of records today; current EV figures represent a lower bound. Coverage improves with additional VIN decoding.

5. Operational Performance

How quickly and reliably a dealer fulfils customer orders from placement to delivery. One of the clearest customer experience differentiators between dealers.

Variable	What it measures	Derived from	Feasibility
Median order-to-delivery days	Middle value of days between customer order and vehicle delivery. Core operational speed metric	Order date + delivery date	Directional
P75 order-to-delivery days	75th percentile OTD — shows what the slower quarter of orders looks like, not just the median	Same	Directional
OTD speed rank within brand	Percentile rank among all dealers for the same brand. 0 = fastest, 1 = slowest	Derived	Directional
On-time delivery rate %	Proportion of deliveries completed on or before the scheduled delivery date. Measures reliability, not just speed	Delivery date vs scheduled date	Directional
On-time rank within brand	Percentile rank on on-time rate vs brand peers	Derived	Directional
OTD consistency score	Standard deviation of OTD days. A dealer with median 60 days but high variance is unreliable even if fast on average	Order + delivery dates	Directional
Estimated delivery accuracy	Average gap between promised delivery date given to customer and actual delivery. Measures how accurately dealers communicate timelines	Estimated vs actual delivery	Directional
Stock turn velocity (yard time)	Median days from vehicle arrival at dealership to customer delivery. Measures how efficiently a dealer moves stock once it arrives	Arrival + delivery dates	Directional
On-time trend	Direction of change in on-time rate vs prior 12 months	Derived by period	Directional

Note: order-to-delivery metrics require order date, which is populated for ~34% of records. All OTD variables should be used for relative comparison within brand peer groups, not absolute benchmarks.

6. Pricing, Revenue & Margin

How a dealer prices vehicles relative to peers — whether they discount heavily to move stock or maintain pricing discipline — and the estimated revenue value flowing through each dealer.

Variable	What it measures	Derived from	Feasibility
Average sale price	Mean actual transaction price per vehicle. Reflects real achieved price not list price	Transaction sale price	Directional
Average discount amount	Mean discount applied at point of sale. Measures how much dealers reduce from list price to close deals	Discount field	Directional
Discount rate %	Discount as a percentage of sale price. Normalised across price segments	Discount / sale price	Directional
Gross margin % (computed)	(Sale price – cost price) / sale price. Measures vehicle-level profit margin per dealer	Sale price + cost price	Directional
Net margin after discounts	Gross margin minus applied discounts. Best single indicator of per-vehicle profitability	Sale + cost + discount	Directional
Aftermarket attach rate	Average aftermarket/accessories revenue per vehicle sold. High attach rate signals strong upsell capability	Aftermarket sale amount	Directional
Dealer options revenue	Average dealer-fitted options revenue per vehicle. Separate from factory options — fully at dealer discretion	Options sale amount	Directional
Estimated revenue potential	Units sold × manufacturer RRP. A 100%-coverage revenue proxy that removes the dependency on actual sale price reporting	Units + RRP table	Ext. data
Average RRP per unit	Mean list price of vehicles this dealer sells. Reflects portfolio price positioning (luxury vs volume dealer)	RRP table by model/year	Ext. data
Price realisation rate	Actual sale price as a percentage of RRP. A dealer achieving 98% of RRP has stronger pricing power than one at 92%	Sale price + RRP table	Ext. data
Discount vs brand average	This dealer's discount rate relative to the brand average. Shows whether a dealer discounts more or less than peers	Derived	Ext. data

Note: transaction sale and cost price are available for approximately 30–33% of records. Margin metrics are directional for that subset. RRP-based variables (marked Ext. data) cover 100% of records once the RRP reference table is available.

7. Dealer Identity & Group Context

Structural context about each dealer — which group they belong to, how large that group is, and whether they operate across multiple brands.

Variable	What it measures	Derived from	Feasibility
Dealer group name	Which automotive group the dealer belongs to (e.g. Eagers Automotive, Servco, Peter Warren). Enables group-level aggregation and benchmarking	Dealer group taxonomy	Ready
Group size (brand count)	Number of brand franchises within the same dealer group. Reflects group scale and negotiating power	Dealer group taxonomy	Ready
Multi-brand dealer flag	Binary: dealer group holds more than one brand franchise	Derived from group size	Ready
Group volume rank	Rank of dealer group by total units delivered nationally. Top 10 groups by volume	Dealer group + delivery records	Ready
Dealer status	Active or inactive. Inactive dealers are typically discontinued brands (e.g. Holden, which exited 2020) and should be excluded from ranking tables	Last delivery date	Ready
Dealer name	Verified trading name of the dealership	Dealer name lookup	Ready
Dealer physical footprint (land size)	Lot size in square metres. Larger footprints generally allow for more stock-on-hand	Cadastral/land registry	Directional

8. Geographic & Competitive Position

Where a dealer sits in the landscape — catchment affluence, how far customers travel, and how much competitive pressure they face from other dealers of the same brand.

Variable	What it measures	Derived from	Feasibility
Metro / regional classification	Whether the dealer is in a major city or regional area. Metro and regional dealers should be benchmarked in separate peer groups	ABS GCCSA lookup (free)	Ext. data
Catchment socio-economic index	SEIFA decile of the dealer's postcode. Explains why some dealers naturally skew toward luxury vehicles and premium pricing	ABS SEIFA lookup (free)	Ext. data
Customer socio-economic index	SEIFA decile of where customers live (delivery postcode). Shows the income profile of the buyer base, independent of the dealer's own location	Delivery postcode + ABS SEIFA	Ext. data
Same-brand dealers within 50 km	Count of competing dealers for the same brand within a 50 km radius. High count = competitive market	Dealer postcode + distance calc	Ext. data
Distance to nearest competitor	Kilometres to the closest dealer selling the same brand. Low distance = high competition; high distance = protected territory	Dealer postcode + distance calc	Ext. data

Variable	What it measures	Derived from	Feasibility
Sole regional dealer flag	Binary: the dealer has no same-brand competitor within 200 km. These dealers have fundamentally different economics from metro dealers	Derived from above	Ext. data
Customer draw radius	Median distance customers travel from their postcode to this dealer. Large radius suggests a destination dealer or a regional area with no closer alternative	Customer delivery postcode	Directional
Catchment population	Estimated population within 25 km of the dealer. Provides market size context for normalising volume metrics	ABS population lookup (free)	Ext. data

Note: geographic enrichment variables marked “Ext. data” use free ABS CSV files (SEIFA, population, GCCSA classification) and a postcode lat/long reference. Distance calculations are pure SQL using the Haversine formula — no third-party API required.

9. Market Share & Competitive Position

How each dealer's volume compares to the total addressable market for their brand in their geography — the most commercially actionable benchmarking dimension.

Variable	What it measures	Derived from	Feasibility
Market share in state	Dealer's units / total new car sales for the same brand in the same state. Direct share of brand volume	VFACTS market data	Ext. data
National brand share	Dealer's units / total national VFACTS registrations for that brand	VFACTS market data	Ext. data
Growth vs market	Dealer's year-on-year growth vs market growth for the same brand and state. Growing faster than the market = gaining share	VFACTS market data	Ext. data
Segment mix vs national average	How this dealer's body style mix compares to the national mix for the same brand. Identifies dealers over- or under-indexed on specific segments	VFACTS + sales data	Ext. data
Geographic white space analysis	Postcodes with high delivery volume but no nearby dealer of the same brand. Identifies territory expansion opportunities for OEMs	Delivery postcode + VFACTS	Ext. data

Note: all market share variables require the VFACTS dataset (monthly new vehicle registrations by make, model, and geography published by the Federal Chamber of Automotive Industries). This dataset is being obtained and will unlock the full market context layer.

10. Extended Analytics — Future Data Sources

The variables below are not computable from the current dataset. Each requires a specific additional data source. They are included here because they represent high-value benchmarking dimensions and should be treated as a product roadmap — connecting each source unlocks a new tier of insights.

Finance & Insurance (F&I System)

Variable	What it measures	Source required	Feasibility
Finance penetration rate %	Proportion of vehicle sales where the customer took dealer-arranged finance. One of the largest revenue streams for dealers beyond the vehicle margin	F&I system integration	Future
Average finance term	Mean loan term in months for financed sales. Longer terms = higher total interest revenue	F&I system	Future

Variable	What it measures	Source required	Feasibility
Insurance attachment rate %	Proportion of sales where the dealer arranged insurance at point of sale. High attachment = strong F&I capability	F&I system	Future
Finance commission per vehicle	Average dealer commission earned on finance products. A key profitability metric not visible in the DMS	F&I system	Future
F&I revenue per unit	Combined finance and insurance revenue per vehicle. Comparable to gross vehicle margin in scale for top-performing F&I dealers	F&I system	Future

Inventory & Stock Management (Live Inventory Feed)

Variable	What it measures	Source required	Feasibility
Current stock on hand	Number of vehicles physically at the dealer awaiting sale. Real-time inventory health	Live inventory system	Future
Aged inventory rate %	Proportion of current stock that has been on the lot for >90 days. High rate = pricing or demand problem	Live inventory system	Future
Days supply	Current stock divided by average daily sales rate. Standard inventory health metric — <30 days = understocked, >90 days = overstocked	Inventory + sales history	Future
Stock-to-sales ratio	Inventory units / monthly sales. Measures how efficiently the dealer converts stock	Inventory + sales history	Future

Customer & Aftersales (CRM / Service History)

Variable	What it measures	Source required	Feasibility
Customer order cancellation rate	Proportion of vehicle orders that were cancelled before delivery. A 15%+ rate signals a process or expectation-setting problem	Order management system	Future
Service retention rate %	Proportion of vehicles sold that return to the same dealer for their first service. Core loyalty metric	Service history / CRM	Future
Workshop utilisation efficiency	Service bay occupancy as a percentage of available capacity. Underutilised workshops represent direct revenue loss	Workshop management system	Future
Customer repeat purchase rate	Proportion of customers who purchase from the same dealer more than once. Hardest metric to compute without customer ID linkage	CRM with customer ID	Future

Variable	What it measures	Source required	Feasibility
Service interval demand forecast	Estimated count of vehicles due for service in coming months, based on delivery date and standard service intervals. Enables workshop capacity planning	Service intervals reference	Future
Net Promoter Score (NPS)	Customer satisfaction score at point of delivery or service. Leading indicator of retention	Survey / CRM	Future

Variable Count Summary

Dimension	Variables proposed	Feasibility
Sales Volume & Growth	7	Ready Full coverage
Channel Mix	5	Directional Directional (65% of records)
Vehicle Portfolio & Segments	10	Ready / Ext. data
Electrification	5	Ready (lower bound today)
Operational Performance	9	Directional Directional (34% of records)
Pricing, Revenue & Margin	11	Directional / Ext. data
Dealer Identity & Group	7	Ready
Geography & Competition	8	Ext. data (free ABS + postcode data)
Market Share	5	Ext. data (VFACTS required)
Core total	67	
Extended — F&I	5	Future
Extended — Inventory	4	Future
Extended — Customer / Aftersales	6	Future
Full roadmap total	82	